

Montrac Robot Operations

Manual

Description of robotic operations and their parameters.

W1/W2 Montrac

V3.0

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OPERATION - ID 10 - MOVE TO HOME

Moves the robot into its home position in PTP motion.

Table of operation parameters			
Index	Name	Values	Description
0	Speed	0 to 100	Percentual speed override of the robot motion (0-100)

Robot actions:

1. The robot moves to its home position.

OPERATION - ID 11 - MOVE LIN CARTESIAN

Moves the robot into a cartesian position using a linear motion.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	B	-360,000 to 360,000	Rotation along Y axis in millidegrees
5	C	-360,000 to 360,000	Rotation along X axis in millidegrees
6	Speed	0 to 100	Percentual speed override of the robot motion
7	Base ID	0 to 127	ID of the base frame in which the motion will be done

Robot actions:

1. The robot moves to the defined position.

OPERATION - ID 12 - MOVE PTP CARTESIAN

Moves the robot into a cartesian position using a PTP motion.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	B	-360,000 to 360,000	Rotation along Y axis in millidegrees
5	C	-360,000 to 360,000	Rotation along X axis in millidegrees
6	Status	0 to 7	KUKA Position Status parameter
7	Turn	0 to 63	KUKA Position Turn parameter
8	Speed	0 to 100	Percentual speed override of the robot motion
9	Base ID	0 to 127	ID of the base frame in which the motion will be done

Robot actions:

1. The robot moves to the defined position.

OPERATION - ID 13 - MOVE PTP AXIS

Moves the robot into an axis specific position using a PTP motion.

Table of operation parameters			
Index	Name	Values	Description
0	A1	-360,000 to 360,000	Axis position millidegrees
1	A2	-360,000 to 360,000	Axis position millidegrees
2	A3	-360,000 to 360,000	Axis position millidegrees
3	A4	-360,000 to 360,000	Axis position millidegrees
4	A5	-360,000 to 360,000	Axis position millidegrees
5	A6	-360,000 to 360,000	Axis position millidegrees
6	Speed	0 to 100	Percentual speed override of the robot motion

Robot actions:

1. The robot moves to the defined position.

OPERATION - ID 20 - PICK

Performs a picking motion with the robot and a gripper. All motions are linear.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	B	-360,000 to 360,000	Rotation along Y axis in millidegrees
5	C	-360,000 to 360,000	Rotation along X axis in millidegrees
6	Speed	0 to 100	Percentual speed override of the robot motion
7	Base ID	0 to 127	ID of the base frame in which the motion will be done
8	Return	0 or 1	If the robot should return to its starting position after the motion
9	Offset	100,000 to 500,000	Offset of the approach position from the provided position in the Z direction in micrometers
10	Gripper Open Position	-100,000 to 100,000	Position of the gripper jaws in micrometers, if allowed by the gripper type

Robot actions:

1. The gripper is moved to the specified position.
2. The robot moves to an approach position, offset from the provided gripping position by the provided offset in the Z direction.
3. The robot moves to the specified gripping position.
4. The gripper is closed.
5. The robot moves back to the approach position.
6. OPTIONAL: The robot returns to its starting position.

OPERATION - ID 21 - PICK VERTICAL

Performs a picking motion with the robot and a gripper. All motions are linear. The gripper is pointing down vertically during the motion, with respect to the base frame. The only rotation which needs to be provided, is along the Z axis.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	Speed	0 to 100	Percentual speed override of the robot motion
5	Base ID	0 to 127	ID of the base frame in which the motion will be done
6	Return	0 or 1	If the robot should return to its starting position after the motion
7	Offset	100,000 to 500,000	Offset of the approach position from the provided position in the Z direction in micrometers
8	Gripper Open Position	-200,000 to 200,000	Position of the gripper jaws in micrometers, if allowed by the gripper type

Robot actions:

1. The gripper is moved to the specified position.
2. The robot moves to an approach position, offset from the provided gripping position by the provided offset in the Z direction.
3. The robot moves to the specified gripping position.
4. The gripper is closed.
5. The robot moves back to the approach position.
6. OPTIONAL: The robot returns to its starting position.

OPERATION - ID 30 - PLACE

Performs a placing motion with the robot and a gripper. All motions are linear.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	B	-360,000 to 360,000	Rotation along Y axis in millidegrees
5	C	-360,000 to 360,000	Rotation along X axis in millidegrees
6	Speed	0 to 100	Percentual speed override of the robot motion
7	Base ID	0 to 127	ID of the base frame in which the motion will be done
8	Return	0 or 1	If the robot should return to its starting position after the motion
9	Offset	100,000 to 500,000	Offset of the approach position from the provided position in the Z direction in micrometers
10	Gripper Open Position	-100,000 to 100,000	Position of the gripper jaws in micrometers, if allowed by the gripper type

Robot actions:

1. The robot moves to a point defined by the specified coordinates, offset by the provided offset in the Z direction.
2. The robot moves directly to the specified cartesian placing point.
3. The gripper is opened to the specified position.
4. The robot moves back to the offset point.
5. OPTIONAL: The robot returns to its starting position.

OPERATION - ID 31 - PLACE VERTICAL

Performs a placing motion with the robot and a gripper. All motions are linear. The gripper is pointing down vertically during the motion, with respect to the base frame. The Only rotation which needs to be provided, is along the Z axis.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	Speed	0 to 100	Percentual speed override of the robot motion
5	Base ID	0 to 127	ID of the base frame in which the motion will be done
6	Return	0 or 1	If the robot should return to its starting position after the motion
7	Offset	100,000 to 500,000	Offset of the approach position from the provided position in the Z direction in micrometers
8	Gripper Open Position	-100,000 to 100,000	Position of the gripper jaws in micrometers, if allowed by the gripper type

Robot actions:

1. The robot moves to a point defined by the specified coordinates, offset by the provided offset in the Z direction.
2. The robot moves directly to the specified cartesian placing point.
3. The gripper is opened to to the specified position.
4. The robot moves back to the offset point.
5. OPTIONAL: The robot returns to its starting position.

OPERATION - ID 40 - APPROACH

Approaches a specified position with the robot. All motions are linear.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	B	-360,000 to 360,000	Rotation along Y axis in millidegrees
5	C	-360,000 to 360,000	Rotation along X axis in millidegrees
6	Speed	0 to 100	Percentual speed override of the robot motion
7	Base ID	0 to 127	ID of the base frame in which the motion will be done
8	Return	0 or 1	If the robot should return to its starting position after the motion
9	Offset	100,000 to 500,000	Offset of the approach position from the provided position in the Z direction in micrometers

Robot actions:

1. The robot moves to an approach position, offset from the provided position by the provided offset in the Z direction.
2. The robot moves to the specified position.
3. The robot moves back to the approach position.
4. OPTIONAL: The robot returns to its starting position.

OPERATION - ID 41 - APPROACH VERTICAL

Approaches a specified position with the robot. All motions are linear. The gripper is pointing down vertically during the motion, with respect to the base frame. The only rotation which needs to be provided, is along the Z axis.

Table of operation parameters			
Index	Name	Values	Description
0	X	-5,000,000 to 5,000,000	X position in micrometers
1	Y	-5,000,000 to 5,000,000	Y position in micrometers
2	Z	-5,000,000 to 5,000,000	Z position in micrometers
3	A	-360,000 to 360,000	Rotation along Z axis in millidegrees
4	Speed	0 to 100	Percentual speed override of the robot motion
5	Base ID	0 to 127	ID of the base frame in which the motion will be done
6	Return	0 or 1	If the robot should return to its starting position after the motion
7	Offset	100,000 to 500,000	Offset of the approach position from the provided position in the Z direction in micrometers

Robot actions:

1. The robot moves to an approach position, offset from the provided position by the provided offset in the Z direction.
2. The robot moves to the specified position.
3. The robot moves back to the approach position.
4. OPTIONAL: The robot returns to its starting position.

OPERATION - ID 50 - BRAKE TEST

Performs brake test on the robot, if it is requested by the robot controller.

Robot actions:

1. The robot performs a brake test.

OPERATION - ID 51 - RUN TOOL COMMAND

Commands the gripper currently attached to the robot to perform to perform a gripper operation, if possible.

Table of operation parameters			
Index	Name	Values	Description
0	Gripper Command	0 to 1000	ID of the requested gripper command

Robot actions:

1. The attached gripper performs the specified operation.

List of available gripper commands:

Gripper Command Number	Name
10	Acknowledge Error
11	Reference
20	Release
21	Grip